Analysis of the Morphometric Characteristics of the Thoracic and Lumbar Pedicles

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A total of 2,905 pedicle measurements were made from T1–L5. Measurements were made from spinal computerized tomography (CT) scan examinations and individual vertebral specimen roentgenograms. Parameters considered were the pedicle isthmus width in the transverse and sagittal planes, pedicle angles in the transverse and sagittal planes, and the depth to the anterior cortex in a line parallel to the midline of the vertebral body and along the pedicle axis. There was no significant difference between data obtained from CT scans and specimen roentgenograms. Pedicles were widest at L5 and narrowest at T5 in the transverse plane. The widest pedicles in the sagittal plane were seen at T11, the narrowest at T1. Due to the oval shape of the pedicle, the sagittal plane width was generally larger than the transverse plane width. The largest pedicle angle in the transverse plane was at L5. The posterolateral to anterolateral pedicle axis orientation in the transverse plane, seen at other levels throughout the thoracolumbar spine, reversed at T12. In the sagittal plane, the pedicles angled caudally at L5 and cephaladly from L3–T1. The depth to the anterior cortex was significantly longer along the pedicle axis than along a line parallel to the midline of the vertebral body at all levels with the exception of T12 and T11. [Key words: pedicles, morphology, thoracolumbar, spine]

INSTRUMENTATION TECHNIQUES OF the spine, using the pedicle as a source of purchase from posterior into the vertebral body, have become an increasingly popular form of spinal fixation. Many systems are currently being used that incorporate plates and screws, rods and screws, and external fixator pins that obtain secure purchase of the vertebra from posterior through the pedicles.[2–10,12] The advantages of these systems are multiple. Fixation and fusion can be limited to fewer segments than previously possible with Harrington rods and segmental intralaminal wire techniques. Laminectomy can be performed at levels to be fused and fusion level need not extend beyond this.

This laboratory experiment was undertaken to gain a detailed knowledge of the pedicle morphology and the depth to which screws may be inserted safely at all levels throughout the thoracolumbar spine.

MATERIALS AND METHODS

Measurements of the morphometric characteristics of the thoracic and lumbar pedicles were obtained in one of two ways. First, postmortem adult spine specimens were disarticulated at individual levels and roentgenograms were obtained of individual vertebra in the transverse plane. In some cases, the entire spine was available for study, while in others, only thoracic or lumbar segments were evaluated. Specimens were then sectioned sagitally in the midline and lateral roentgenograms of the hemisectioned vertebra were obtained. The roentgenographic technique was kept constant. There was no detectable magnification with this technique. Morphologic measurements were obtained directly from the x-ray films. Second, computerized axial tomographic (CAT) examinations were reviewed and morphometric measurements were made using the computer software. The most commonly studied region was the lumbar spine, L3–S1.

The narrowest dimension in both the transverse and sagittal planes was chosen as the transverse pedicle isthmus width and the sagittal pedicle isthmus width, respectively. Transverse pedicle angles were obtained by measuring the angle between a line perpendicular to the transverse isthmus and a line parallel to the vertebral midline in the transverse plane. Sagittal pedicle angles were measured between a line perpendicular to the sagittal isthmus and the posterior vertebral body border in the sagittal plane.

Measurements of the depth to anterior cortex, defined as the distance from the most posterior aspect of the transverse process to the anterior cortex, were obtained in two ways. The measurements were taken in the line of the transverse pedicle angle and in a line parallel to the midline of the vertebral body.

RESULTS

CT Scan Versus Specimen Roentgenograms

There was no significant difference between measurements made from the CT scans and the specimen roentgenograms ($P < 0.05$).

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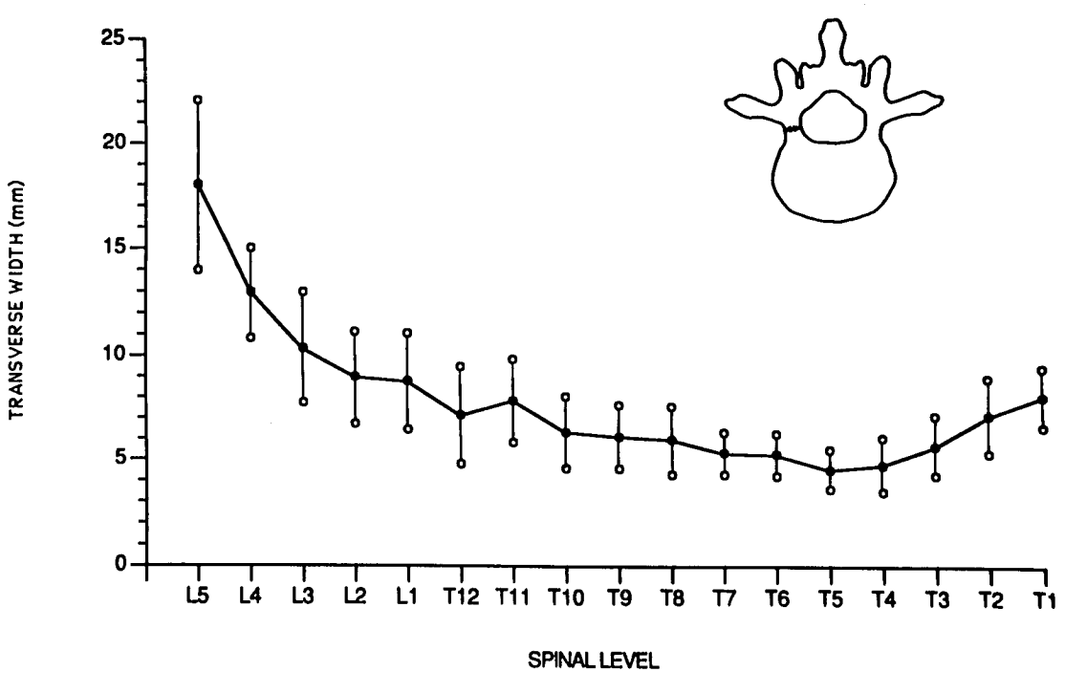


Fig 1. Transverse pedicle isthmus widths.

Pedicle Isthmus Width

Transverse: The widest pedicle was seen at the L5 level with a mean of 18.0 mm (range: 9.1–29.0 mm). The narrowest was seen at the T5 level with a 4.5-mm mean width (range: 3.0–7.0). The narrowest width in the lumbar spine was seen at L1 with a mean of 8.7 mm (range: 4.5–13.0 (Figure 1, Table 1).

Sagittal: The widest sagittal pedicle isthmus width was seen at T11 with a mean value of 17.4 mm (range: 12.5–24.1). The narrowest was seen at T1 with a mean of 9.9 mm (range: 7.0–14.5). In the lumbar spine, the narrowest width in this plane was seen at L5, 14.0 mm (range: 9.5–19.0) and the widest at L1, 15.4 mm (range: 11.0–21.0) (Figure 2, Table 2).

Pedicle Angle

Transverse: The largest angle from the midline was seen at L5 with the mean pedicle of 29.8° (range: 19.0–44.0°). The shallowest angle was seen at T12 and this was −4.2° (range: −17.0–+14.5°).

Table 1. Transverse Pedicle Isthmus Width (mm)

Level	*N*	*Mean*	*Range*	*SD*	*SEM*
L5	56	18.0	9.1–29.0	4.1	0.5
L4	36	12.9	9.1–17.0	2.1	0.4
L3	49	10.3	5.3–16.0	2.6	0.4
L2	30	8.9	4.0–13.0	2.2	0.4
L1	26	8.7	4.5–13.0	2.3	0.5
T12	36	7.1	3.0–11.0	2.3	0.4
T11	27	7.8	3.1–12.0	2.0	0.4
T10	24	6.3	3.1–8.5	1.7	0.3
T9	27	6.1	3.7–9.0	1.5	0.3
T8	41	5.9	2.5–9.0	1.6	0.3
T7	32	5.3	3.2–7.0	1.0	0.2
T6	33	5.2	3.0–6.5	1.0	0.2
T5	34	4.5	3.0–7.0	0.9	0.2
T4	25	4.7	2.5–7.0	1.3	0.3
T3	27	5.6	3.0–8.5	1.4	0.3
T2	24	7.0	3.5–10.0	1.8	0.4
T1	24	7.9	5.0–10.0	1.4	0.3

Number of specimen roentgenograms = 387.
Number of specimen CT scans = 160.
Total studies = 547.

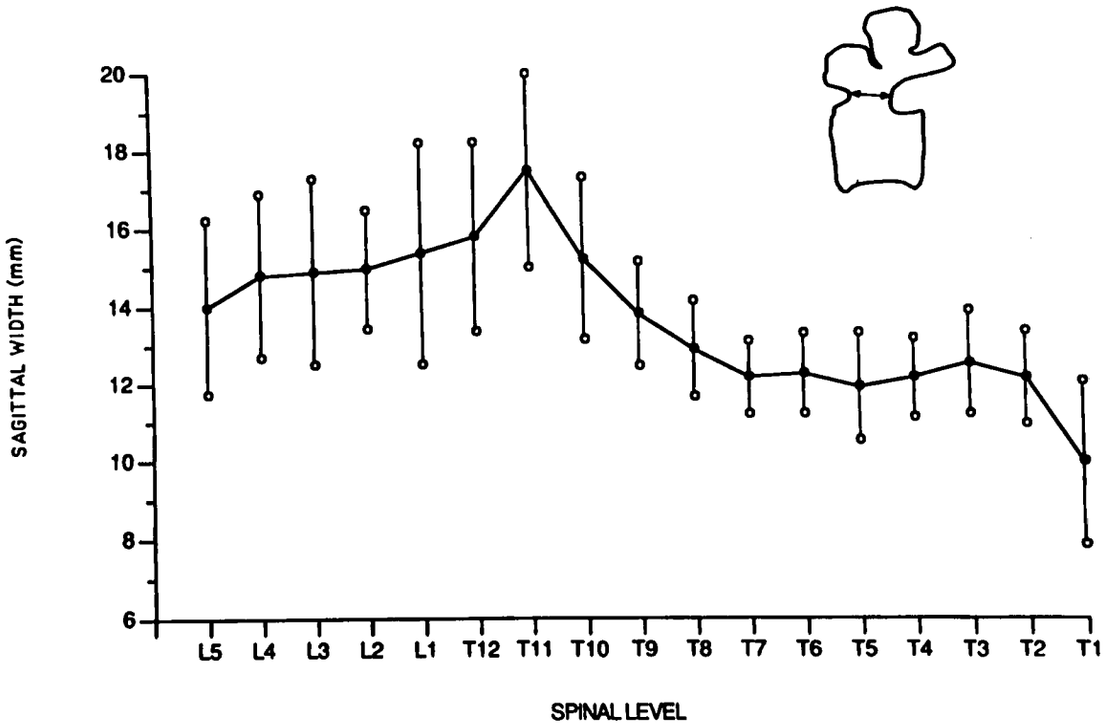


Fig 2. Sagittal pedicle isthmus widths.

In the lumbar spine the shallowest angle was 10.9° (range: 6.5–14.5°) at L1. The largest angle of the thoracic spine was T1 with a mean value of 26.6° (range 16.0–33.5°) (Figure 3, Table 3).

Sagittal: The largest cephalad angulation was seen at T2 with a mean value of 17.5° (range: 7.0–32.0°). The L5 pedicle angled caudally −1.8° (range: −8.0–+8.0°). The pedicle angle in this plane was essentially perpendicular to the posterior vertebral margin in the remainder of the lumbar vertebrae. The thoracic vertebral pedicles all appeared to angle in the cephalad direction (Figure 4, Table 4).

Depth to Anterior Cortex

Through pedicle axis: The shortest distance to the anterior cortex was seen at T1 with a mean of 36.9 mm (range: 26.0–52.0 mm). The longest distance to the anterior cortex was seen at L2 and L3,

Table 2. Sagittal Pedicle Isthmus Width (mm)

Level	*N*	*Mean*	*Range*	*SD*	*SEM*
L5	52	14.0	9.5–19.0	2.3	0.3
L4	51	14.8	8.9–18.6	2.1	0.3
L3	27	14.9	8.3–18.2	2.4	0.5
L2	23	15.0	10.0–18.0	1.5	0.3
L1	21	15.4	11.0–21.0	2.8	0.6
T12	24	15.8	12.0–20.0	2.4	0.5
T11	30	17.4	12.5–24.1	2.5	0.5
T10	29	15.2	11.0–21.6	2.0	0.4
T9	32	13.8	10.6–15.5	1.3	0.2
T8	43	12.8	10.5–15.7	1.2	0.2
T7	39	12.1	10.5–14.4	1.0	0.2
T6	44	12.2	9.4–14.0	1.0	0.2
T5	42	11.9	6.5–13.5	1.4	0.2
T4	38	12.1	9.1–13.5	1.0	0.2
T3	40	12.4	10.0–15.0	1.3	0.2
T2	32	12.0	10.0–14.0	1.2	0.2
T1	28	9.9	7.0–14.5	2.0	0.4

Number of specimen roentgenograms = 433.
Number of specimen CT scans = 162.
Total studies = 595.

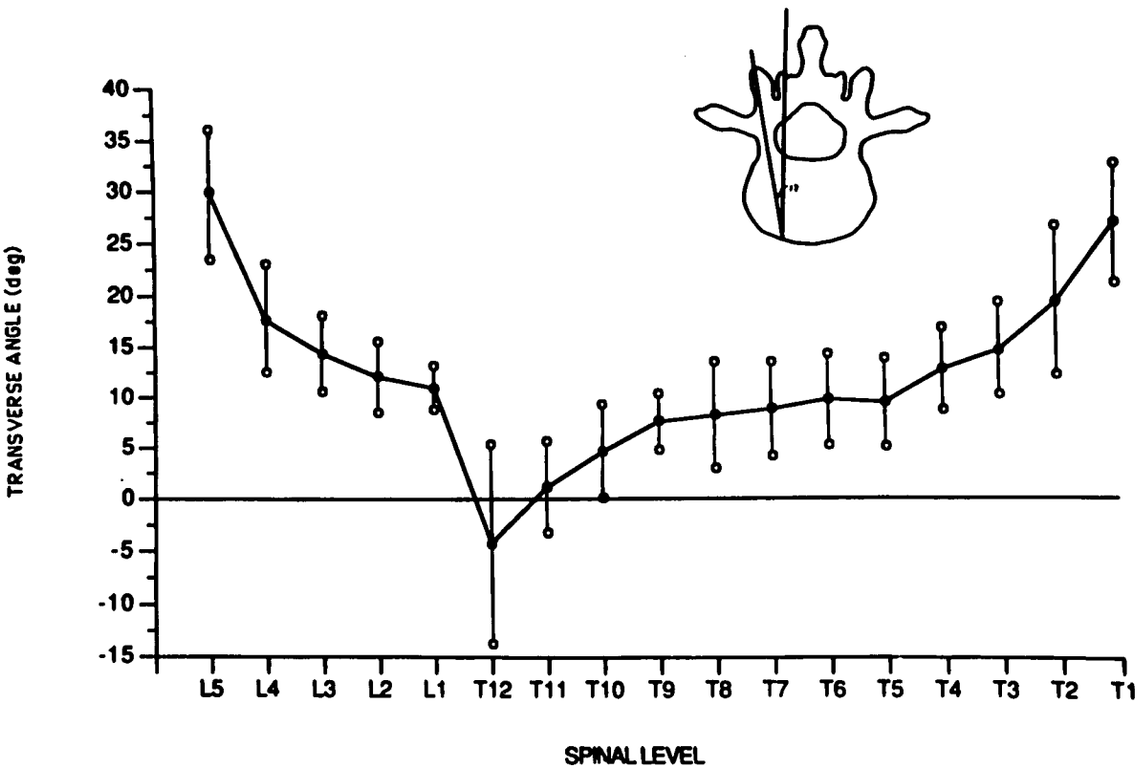


Fig 3. Transverse pedicle angles.

51.9 mm (ranges: 45.0–58.0 and 42.0–62.0 mm, respectively) (Figure 5, Table 5).

Parallel to the midline axis of vertebral body: The shortest distance to the anterior cortex was seen at T1, 26.7 mm (range: 17.0–43.0 mm). The longest distance was at L2, 45.5 mm (range: 37.0–52.0 mm).

In all cases except at the T11 and T12 levels, there was a statistically significant difference in the distance to the anterior cortex between the line drawn parallel to the midline axes of the vertebral body and line along the pedicle axis ($P < 0.05$) (Figure 5, Table 5).

DISCUSSION

A detailed knowledge of pedicle size and dimensions is crucial when using the pedicle to gain purchase of the vertebra. Screw sizes must take pedicle dimensions into consideration.

Pedicle dimensions have been studied by other authors. Saillant

Table 3. Transverse Pedicle Angle (°)

Level	*N*	*Mean*	*Range*	*SD*	*SEM*
L5	76	29.8	19.0–44.0	6.3	0.7
L4	76	17.7	5.5–27.5	5.2	0.6
L3	68	14.4	8.0–23.5	3.8	0.5
L2	38	12.0	5.0–17.5	3.5	0.6
L1	26	10.9	6.5–14.5	2.2	0.4
T12	24	−4.2	−17.0–14.5	9.5	1.9
T11	24	1.2	−6.0–9.5	4.4	0.9
T10	20	4.6	0.0–13.5	4.6	1.0
T9	30	7.6	0.0–10.5	2.8	0.5
T8	42	8.1	−3.0–18.5	5.2	0.8
T7	32	8.7	−3.5–18.0	4.7	0.8
T6	30	9.6	−2.0–17.5	4.5	0.8
T5	36	9.4	2.0–18.5	4.4	0.7
T4	28	12.6	7.0–22.5	4.1	0.8
T3	28	14.6	7.5–21.5	4.4	0.8
T2	26	19.1	6.0–32.5	7.1	1.4
T1	24	26.6	16.0–33.5	5.6	1.1

Number of specimen roentgenograms = 410.
Number of specimen CT scans = 218.
Total studies = 628.

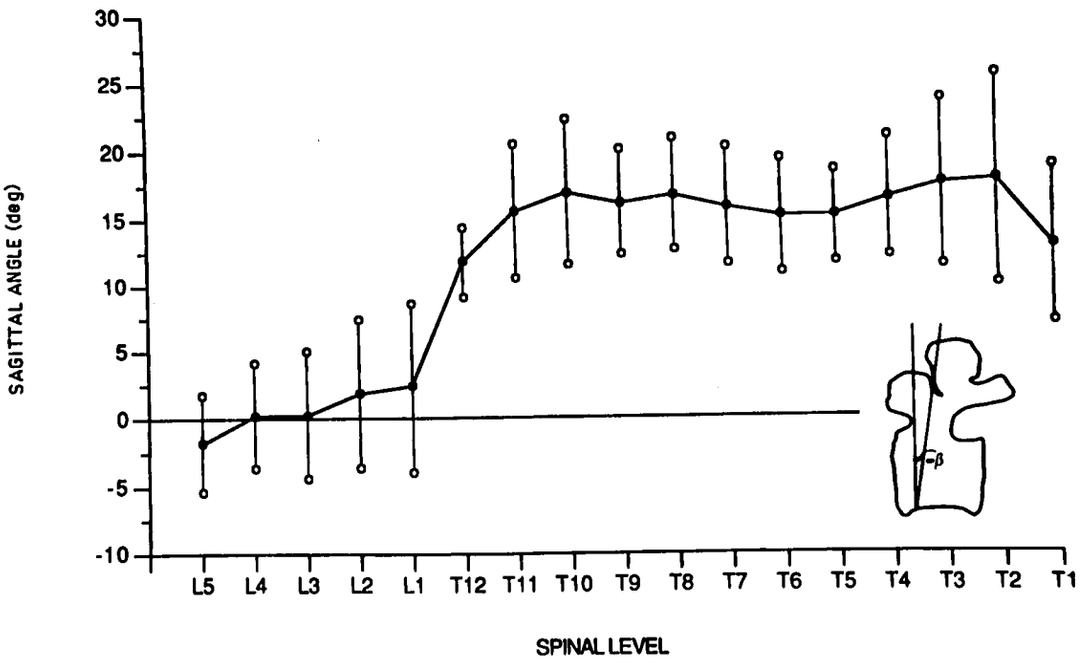


Fig 4. Sagittal pedicle angles.

measured the thoracic and lumbar pedicles, but only included mean values.[11] Krag et al, using CT scan data, measured the lower thoracic and lumbar spine.[5] Berry et al, using a direct specimen measurement technique, looked at select pedicles in the spine.[1] In the current study, every level of the thoracolumbar spine was analyzed. In addition to presenting mean values and ranges, statistical evaluations were performed and the depth in the anterior cortex was measured and statistically evaluated at all levels. The data presented here is in close agreement with the existing data of others.[1,5,11]

The oval-shaped pedicle is narrowest in the transverse plane. It is in this plane that the maximum allowable pedicle screw diameter is determined. Screws larger than the pedicle can cause pedicle breakage or thread cutout. Insertion of methyl methacrylate can be disastrous if it leaks through a defect created in the pedicle by the screw.[13,14] The pedicle is much wider in the sagittal plane and does not limit the maximum diameter of a pedicle screw (Figures 1 and 2).

The transverse plane pedicle angle was medial in the lumbar spine and gradually returned to neutral by L1 and then began to angle laterally at T12. The angles then gradually increase to a maximum medial angulation in the thoracic spine at T1 (Figure 3). In the sagittal plane the pedicles angle in an increasingly cephalad direction from the lumbar spine through the thoracic spine (Figure 4).

The distance to the anterior cortex is important in preventing

Table 4. Sagittal Pedicle Angle (°)

Level	*N*	*Mean*	*Range*	*SD*	*SEM*
L5	47	−1.8	−8.0-8.0	3.5	0.5
L4	52	0.2	−6.0-7.0	3.9	0.5
L3	28	0.2	−10.0-12.0	4.7	0.9
L2	20	1.8	−10.0-13.0	5.5	1.2
L1	22	2.4	−13.0-15.0	6.3	1.3
T12	20	11.6	−7.0-11.0	2.6	0.6
T11	27	15.4	−6.0-24.0	4.9	0.9
T10	26	16.8	8.0-29.0	5.3	1.0
T9	30	16.0	9.0-14.0	3.8	0.7
T8	33	16.6	8.0-25.0	4.1	0.7
T7	36	15.7	6.0-22.0	4.2	0.7
T6	40	15.0	7.0-25.0	4.2	0.6
T5	28	15.0	7.0-20.0	3.4	0.6
T4	33	16.3	8.0-25.0	4.4	0.8
T3	34	17.3	8.0-31.0	6.0	1.0
T2	25	17.5	7.0-32.0	7.7	1.5
T1	21	12.6	4.0-25.0	5.8	1.3

Number of specimen roentgenograms = 372.
Number of specimen CT scans = 150.
Total studies = 522.

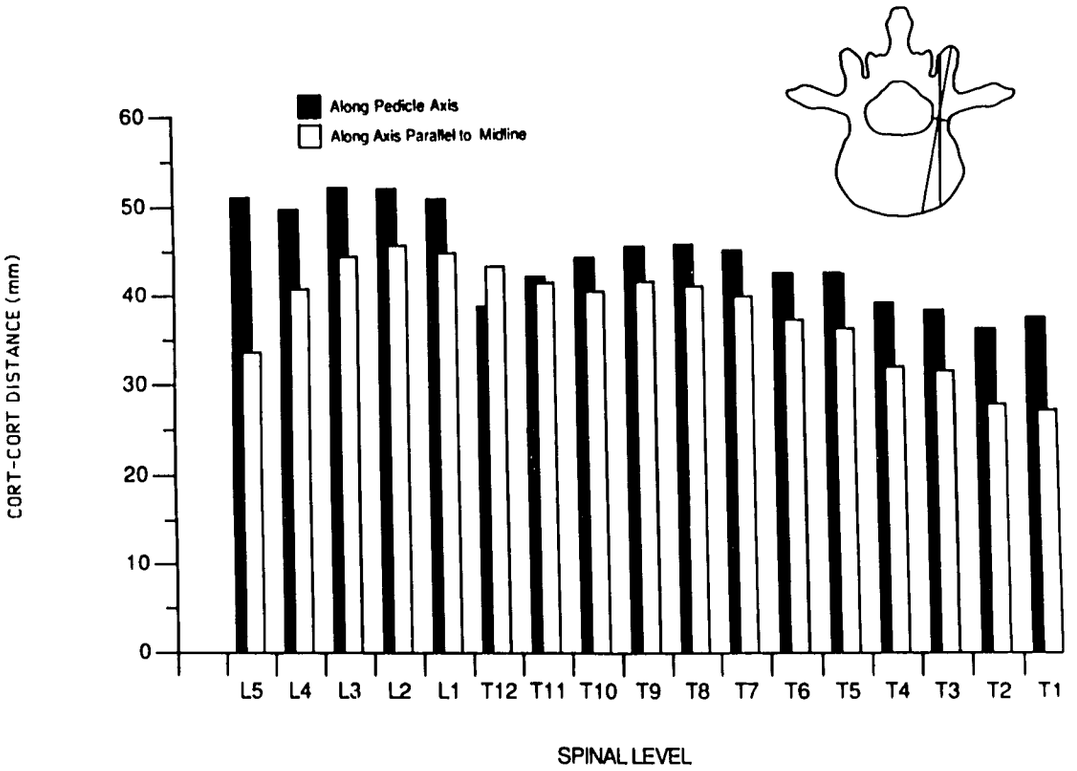


Fig 5. Distance to the anterior cortex through pedicle angle axis versus through line parallel to midline axis of vertebra.

anterior cortex perforation and possible injury to vital structures. The distance to the anterior cortex is significantly greatest along the pedicle axis at all levels except T11 and T12 ($P < 0.05$) (Figure 5). Many authors have recommended straight anterior pedicle screw insertion, ie, axis parallel to midline. However, in the laboratory, the deeper-inserted screws withstood the largest number of cycles in the cephalad-caudad and medial-lateral directions before failure, and screws purchasing the anterior cortex had greater pullout strengths than screws inserted to a lesser depth.[5,13]

In addition, the radiographic blind spot encountered intraoperatively during screw insertion by use of lateral fluoroscopy, decreases in size the closer the screw tip comes to the vertebral body midline as it advances toward the anterior cortex. Thus, there is less chance of inadvertent anterior cortex penetration with screws inserted along the pedicle axis as opposed to parallel to the midline axis of the vertebra at all levels, with the exception of T11 and T12.

In summary, pedicle dimensions and angles change throughout the spine. A detailed knowledge of these relationships is important

Table 5. Depth to Anterior Cortex: Through Pedicle Axis Versus Through Parallel to Midline Axis of the Vertebra

		Pedicle Axis				*Midline Axis*			
Level	*N*	*Mean*	*Range*	*SD*	*SEM*	*Mean*	*Range*	*SD*	*SEM*
L5	69	51.0	42-62	4.7	0.6	33.7	20-44	5.6	0.7
L4	74	49.7	43-62	4.4	0.5	40.7	27-48	4.3	0.5
L3	64	51.9	42-62	4.6	0.6	44.4	31-54	5.1	0.6
L2	35	51.9	45-58	3.7	0.6	45.5	37-52	3.7	0.6
L1	26	50.7	42-57	4.3	0.8	44.7	36-51	4.5	0.9
T12	24	38.6	23-61	11.0	2.2	43.2	34-53	5.0	1.0
T11	29	41.8	24-56	6.5	1.2	41.2	32-49	3.8	0.7
T10	24	44.0	36-55	5.0	1.0	40.1	29-50	4.6	0.9
T9	30	45.2	35-56	4.9	0.9	41.2	31-52	5.3	1.0
T8	42	45.4	37-57	4.8	0.7	40.6	31-51	4.1	0.6
T7	32	44.6	38-54	3.7	0.7	39.5	31-48	3.7	0.7
T6	28	42.1	34-51	4.7	0.9	36.9	26-46	5.0	0.9
T5	36	41.9	33-51	4.7	0.8	35.9	26-43	4.1	0.7
T4	28	38.5	33-49	3.9	0.7	31.5	25-39	3.2	0.6
T3	26	37.7	29-47	5.1	1.0	31.1	25-38	3.1	0.6
T2	24	35.7	26-45	4.7	1.0	27.4	15-39	6.3	1.3
T1	22	36.9	26-52	5.8	1.2	26.7	17-43	5.8	1.2

Number of specimen roentgenograms = 403.
Number of specimen CT scans = 210.
Total studies = 613.

to any surgeon contemplating the use of the pedicle for screw purchase to prevent screw cutout and failure of fixation or neurologic injury. A knowledge of the depth to the anterior cortex is essential in choosing screw lengths but should always be checked intraoperatively with fluoroscopic control during screw insertion.

REFERENCES

1. Berry JL, Moran JM, Berg WS, Steffee AD: A morphometric study of human lumbar and select thoracic vertebra. Spine (In press)
2. Edwards CC: Sacral fixation device, design and preliminary results. Presented at the 19th Annual Meeting of the Scoliosis Research Society, Orlando, Florida, September 19–22, 1984, p 135
3. Field BT, Wiltse LL, Zindrick MR, Widell EH, Thomas JC: The Long Beach spinal fixation system. Presented at the 2nd Annual Meeting of the North American Lumbar Spine Association, Laguna Niguel, California, July 27, 1985
4. Kostuik JP, Errico TJ, Gleason TF: Techniques of internal fixation for degenerative conditions of the spine. Clin Orthop 203:219, 1986
5. Krag MH, Beynnon BO, Pope MH, Frymoyer JW, Haugh LD: An internal fixation for posterior application to short segments for the thoracic, lumbar and lumbosacral spine: Design and testing. Clin Orthop 203:75, 1986
6. Louis R: Single staged posterior lumbosacral fusion by internal fixation with screw plates. Presented at the Annual Meeting of the International Society for the Study of the Lumbar Spine, Sidney, Australia, 1985, p 46
7. Luque E: Intrapeduncular segmental fixation. Clin Orthop 203:126, 1986
8. Magerl FP: External skeletal fixation of the lower thoracic and the lumbar spine. In Current Concepts of External Fixation of Fracture. Edited by Uhthoff HK, Stahl E. New York, Springer–Verlag, 1982
9. Magerl FP: Stabilization of the lower thoracic and lumbar spine with external skeletal fixation. Clin Orthop 189:125, 1984
10. Roy-Camille R, Saillant G, Berteaux D, Marie-Anne S, Mamoudy P: Vertebral osteosynthesis using metal plates: Its different uses. Chirurgie 105:597, 1979
11. Saillant G: Anatomic study of vertebral pedicles, surgical application (in French) Rev Chir Orthop 62:151, 1976
12. Steffe A, Biscup R, Sitkowski D: Segmental spine plate with pedicle screw fixation: A new internal fixation device for disorder of the lumbar and thoracic spine (unpublished manuscript)
13. Zindrick MR, Wiltse LL, Widell EH, et al: A biomechanical study of intrapeduncular screw fixation in the lumbosacral spine. Clin Orthop 203:99–111, 1986
14. Zindrick MR, Patwardhen AG, Lorenz M, et al: The effect of methylmethacrylate augmentation upon pedicle screw fixation in the spine. Presented at the Annual Meeting of the International Society for the Study of the Lumbar Spine, Dallas, Texas, May 1986 (Abstract)

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